LEARNER GUIDE

Shapes that hold their form

Constructions, Loci and Bearings · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Look at a bridge, a roof or a crane and you keep seeing the **triangle**. That is no accident. A triangle built from three fixed-length bars *cannot change its shape* — it is **rigid**. A four-sided frame with the same fixed bars can still lean and squash. In this lesson you construct triangles and quadrilaterals for yourself and discover which shapes their sides fix — and how engineers use a single diagonal to lock a floppy quadrilateral solid.

What you are learning to notice

- Three fixed side lengths **fix a triangle completely** (this is SSS, from Lesson 1) — there is no other shape it can take.
- A quadrilateral with four fixed sides can still **flex** into many shapes: four sides do not fix it.
- A **diagonal** splits a quadrilateral into two triangles; because each triangle is rigid, the whole frame becomes rigid.
- This is **triangulation** — the reason diagonals appear in gates, shelves, bridges and pylons.

Moving through it, one step at a time

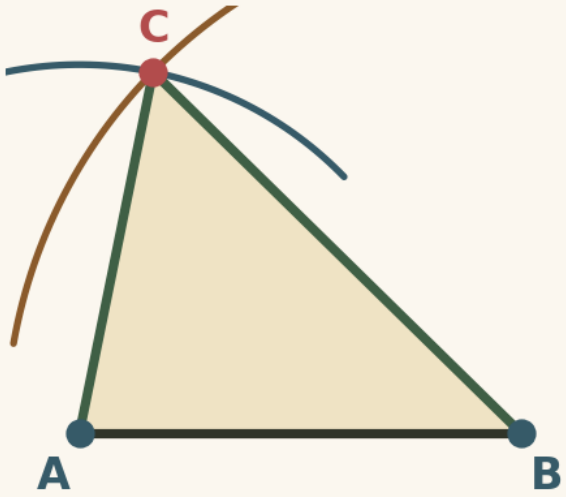
Construct a triangle from three sides exactly as in Lesson 1: draw a base, swing an arc of the right length from each end, and join the crossing. To build a quadrilateral, construct its four sides in turn. To make a quadrilateral rigid, add one diagonal from a corner to the opposite corner — it splits into two triangles and can no longer flex. Leave your construction arcs showing.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Construct a triangle from three sides

Construct a triangle with sides 6 cm, 5 cm and 7 cm — the single shape those three sides fix.

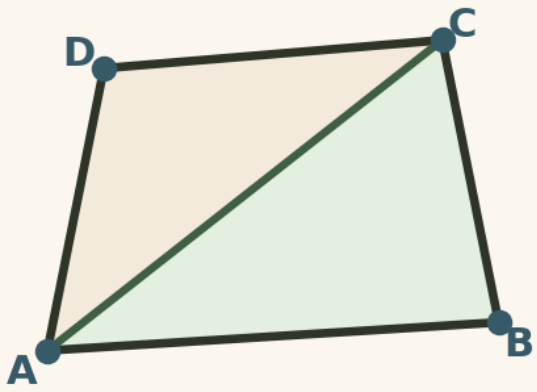


1. Draw the base AB = 6 cm.	AB = 6 cm
2. Swing an arc from A, radius 5 cm.	radius = 5 cm
3. Swing an arc from B, radius 7 cm.	radius = 7 cm
4. Mark C where the arcs cross; join A–C and B–C.	join A–C, B–C

There is only one shape of triangle with sides 6, 5, 7 (SSS): the three sides fix it completely.

Example 2 — Brace a quadrilateral to make it rigid

A quadrilateral ABCD can flex even with fixed sides. Add one diagonal to make it rigid.



1. Construct the quadrilateral ABCD (its four sides).	4 sides
2. Draw the diagonal from A to C.	join A–C
3. The diagonal splits ABCD into triangles ABC and ACD.	2 triangles
4. Each triangle is rigid, so the whole frame is now rigid.	rigid

One diagonal triangulates the quadrilateral: two rigid triangles lock the shape.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (constructing with compass arcs (Lesson 1), and what a triangle and quadrilateral are), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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